IPv6, And What It Means For The Future

IPv6 is the next generation Internet protocol, designed by the Internet Engineering Task Force, to replace the current version (IPv4) of the Internet Protocol. IPv4, which most of the Internet uses, has been around for 20 years. Despite being remarkably resilient, cracks are beginning to show up—most noticeably the growing shortage of addresses; there are four billion possible. This may seem like a lot, but at the rate the Internet is expanding, four billion will become too little.

IP addresses will soon trickle down to everything—the easiest example being handhelds, and esoteric ones like dog collars. The larger address space provided by IPv6 will soon become necessary. IPv6 supports autoconfiguration, so your phone or dog collar can be configured without your having to do much. Besides, IPv6 makes for a hierarchical addressing system: so, for example, your home could have a top-level address, the rooms in it could have sub-addresses, each device in each room sub-sub-addresses, and so on.

IPv6 fixes a number of problems with IPv4; apart from increasing the number of available addresses, it adds improvements in areas such as routing. IPv6 is expected to gradually replace IPv4; the two will co-exist for some years during the transition period.

An IPv6 address is 128 bits long, unlike today's 32-bit addresses. It uses hexadecimal digits. A typical IPv6 address looks something like this: 4FDE:0000:0000:0002:0022:F376:FF3B:AB3F.

Is I2 Any Better?

In a world connected by an I2-like network, security concerns would be higher than what they are today.

Is security inbuilt in I2? Is there an emphasis on security? Speaking in an interview with CREN (Center for Research and Educational Networking) TechTalk, Ted Hanss, director of applications development at I2 said, "I would say that we are not really trying to do anything different, as much as we are trying to ensure that security is part of the environment. We have demonstrated that you can, in real-time, encrypt a video screen—full-motion, full-screen video—without interfering and putting in so much delay that it becomes unusable. What we are really doing is finding a way to make sure that security features are available to application developers."

Encryption is of the essence—IPv6 mandates the IPsec protocol, short for IP Security. It is a set of protocols to support secure exchange of packets at the IP layer.

Security will obviously be one of the biggest issues to face I2 in the years to come. In fact, the first well-documented DDoS (Distributed Denial of Service) attack occurred in August of 1999, when a DDoS tool was deployed on at least 227 systems, of which at least 114 were on Internet2. All this just to flood a single computer at the University of Minnesota!

The Internet2 environment, with its solid, clean interface, is an ideal testbed for a lot of things, and is a crucible of innovation.

As Hanss puts it: "Application developers with security features available to them will surely innovate—and these innovations should trickle down to the rest of the Internet."

With Internet2, we can hope for a cleaner, more secure Internet of the future—not to forget the blazingly fast speeds! ⊠

ram_mohan@thinkdigit.com